Day 2 morning — learning a proportion

Bayesian statistics for PhD researchers · Lecturer: Lu Cheng

Tuesday 10 November 2026 · Kroghstræde 3, room 4.110

Likelihood · beta distribution · beta–binomial update · posterior summaries · prior sensitivity · first look at Stan

Today

08:45	**Lecture A**	Proportions: beta–binomial updating · first look at Stan
09:45	**Exercise A**	Your own update in R, two priors, a grid check · question set
11:10	**Peer review A**	Partner interviews · plenary · everyone back at 11:10
11:55	**Lunch**	
12:30	**Lecture B**	Prediction · counts with exposure · map of distributions
13:30	**Exercise B**	Posterior predictive counts and one predictive plot · question set
15:00	**Peer review B**	Partner interviews · plenary · everyone back at 15:00
15:45	**Daily diary**	Mentimeter: 3 things learned, 3 not yet familiar

By the end of this lecture you can

1. **Write** the likelihood for y successes in n trials
2. **Update** a Beta prior to a Beta posterior by hand
3. **Summarise** a posterior: mean, interval, P(θ > c)
4. **Judge** how much the prior matters
5. **Recognise** the same model written in Stan

From five bags to any θ

Day 1: the marble game

θ could be one of five values (five bags).

prior × likelihood, then divide by the sum.

A table with five rows.

→

Today: θ is any value in [0, 1]

θ = probability that a bus departure is on time.

prior × likelihood, then divide by an integral.

A curve instead of a table.

Today's example (invented data)

A city bus line gets a new timetable. We watch the first n = 20 departures: y = 14 leave on time.

What can we say about θ?

Bayes' rule for a parameter

$$p(\theta \mid y) = p(y \mid \theta) \cdot p(\theta) \,/\, p(y)$$

posterior
what we believe about θ after the data

likelihood
how probable the data are for each θ

prior
what we believed before the data

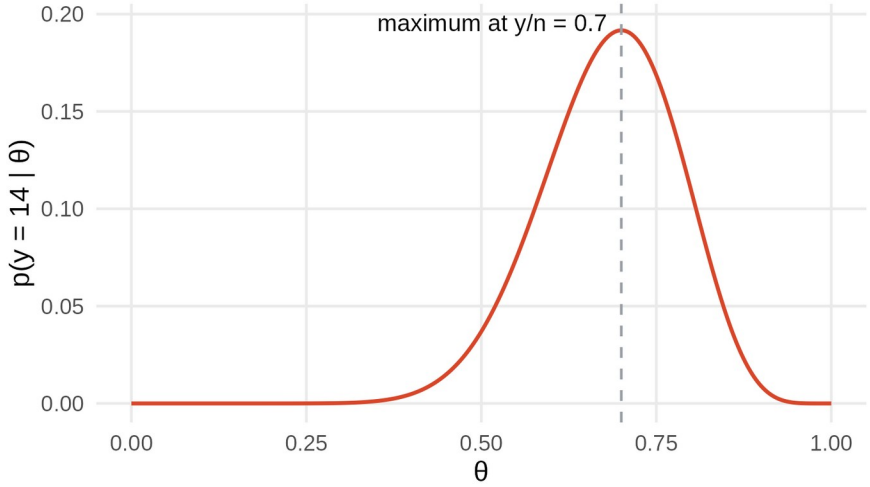
evidence
$p(y) = \int p(y \mid \theta)\, p(\theta)\, d\theta$ — a constant that makes the area 1

Working rule: posterior ∝ likelihood × prior. The constant can wait.

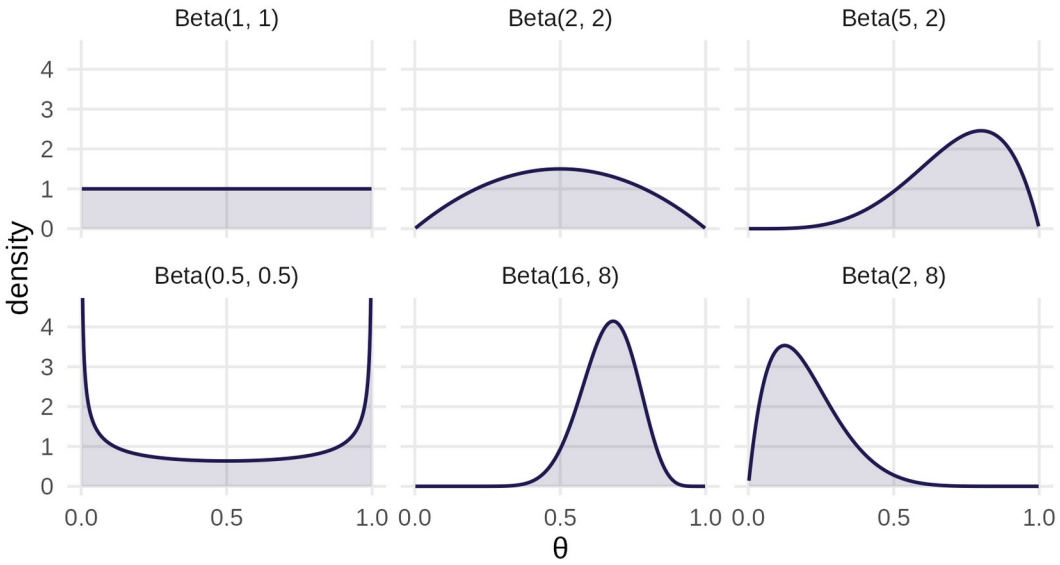
The likelihood: binomial in y, a curve in θ

$$p(y \mid \theta) = C(n, y) \cdot \theta^{y} (1 - \theta)^{n-y}$$

- Fix the data (y = 14, n = 20) and vary θ.
- Highest at θ = y/n = 0.7.
- It is not a distribution for θ: the area under it is not 1.
- Only its shape in θ matters; C(n, y) does not depend on θ.



The beta distribution: a prior for a proportion



- θ ~ Beta(a, b) on [0, 1]
- density $\propto \theta^{a-1}(1-\theta)^{b-1}$
- mean a / (a + b)
- read a and b as earlier successes and failures
- a + b: how much the prior weighs
  - Beta(1, 1) = uniform
  - R: dbeta, pbeta, qbeta, rbeta

The beta–binomial update

$$p(\theta \mid y) \propto \theta^{y}(1-\theta)^{n-y} \times \theta^{a-1}(1-\theta)^{b-1}$$
$$= \theta^{a+y-1}(1-\theta)^{b+n-y-1}$$

$$\theta \mid y \sim \text{Beta}(a + y,\ b + n - y)$$

Add successes to a

14 on-time departures: a = 2 → 16

Add failures to b

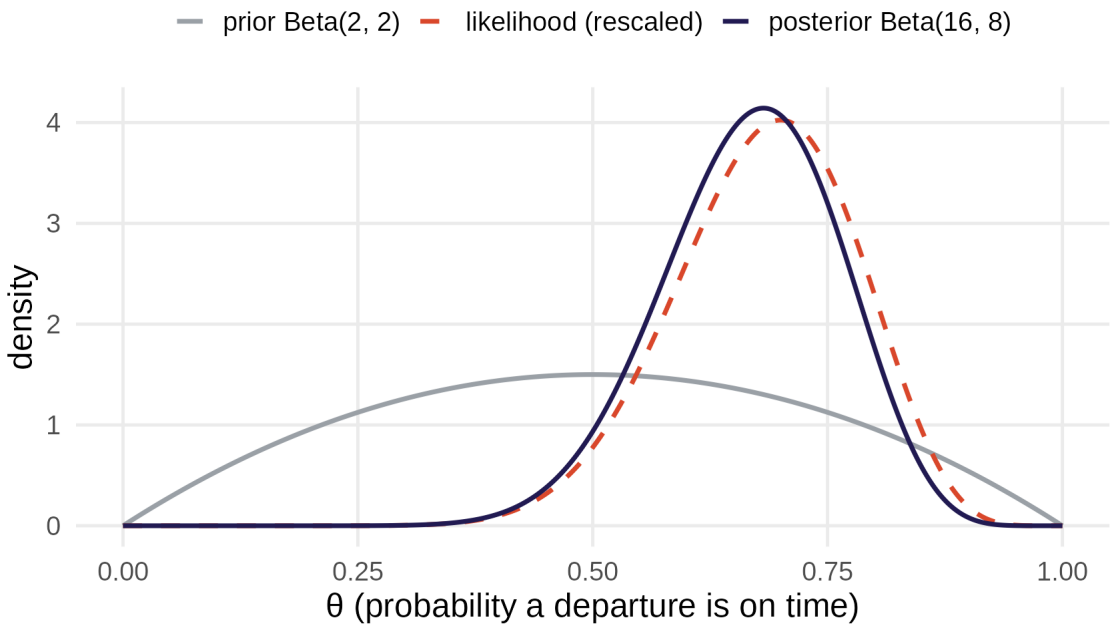
6 late departures: b = 2 → 8

Conjugate prior

Beta in, Beta out — the posterior stays in the prior's family

This derivation is the one you do in full in Exercise A.

Prior × likelihood → posterior



- Prior Beta(2, 2): mild, centred at 0.5
- Data: 14 of 20
- Posterior Beta(16, 8)
- The posterior sits between prior and data, closer to the data

The posterior mean is a compromise

$$E[\theta \mid y] = (a + y) / (a + b + n) = w \cdot a/(a + b) + (1 - w) \cdot y/n, \quad w = (a + b)/(a + b + n)$$

0.50	0.70	0.17	0.667
prior mean	data y/n	weight on prior, w = 4/24	posterior mean

The prior counts as a + b extra observations. More data → less weight on the prior.

Summarising the posterior Beta(16, 8)

```
a_post <- 16; b_post <- 8

a_post / (a_post + b_post)      # mean   0.667
qbeta(0.5, a_post, b_post)      # median 0.671
qbeta(c(0.05, 0.95), 16, 8)     # 90%: 0.504, 0.814
1 - pbeta(0.6, a_post, b_post)  # P(θ > 0.6) = 0.763
```

In words

Given the prior and these 20 departures, there is a 90% probability that θ lies between 0.50 and 0.81, and a 76% probability that more than 60% of departures are on time.

- Report a centre, an interval and the probability that answers the question.
- Equal-tailed interval: 5% of posterior probability on each side.

What a 90% interval says — and does not say

Credible interval (Bayesian)

“Given the model, prior and data, θ is in [0.50, 0.81] with probability 0.9.”

A statement about θ, for these data.

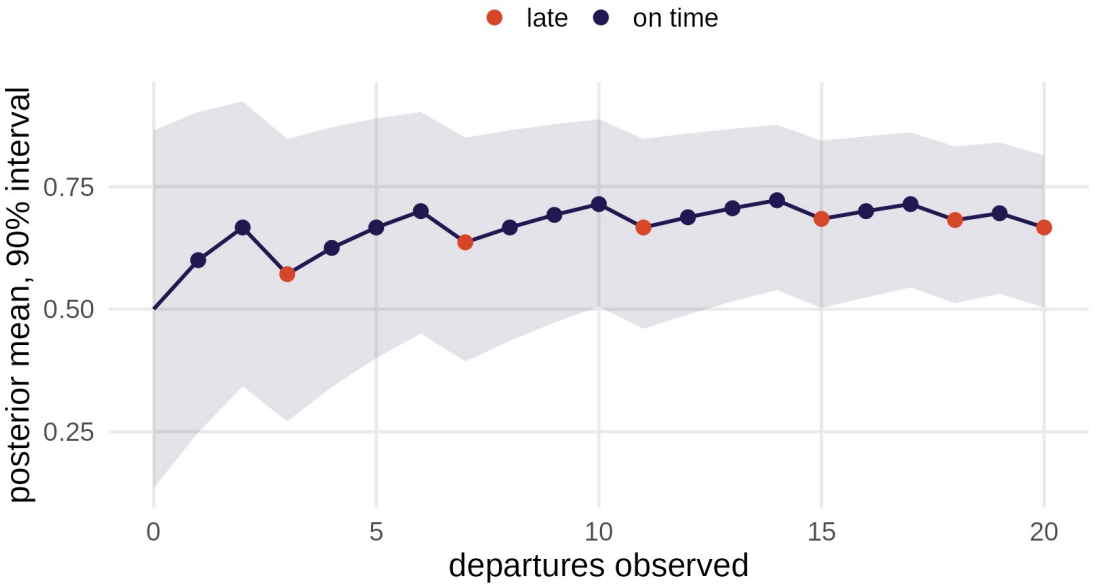
Confidence interval (frequentist)

“The procedure covers the true θ in 90% of repeated samples.”

A statement about the procedure. Any one interval either covers θ or not.

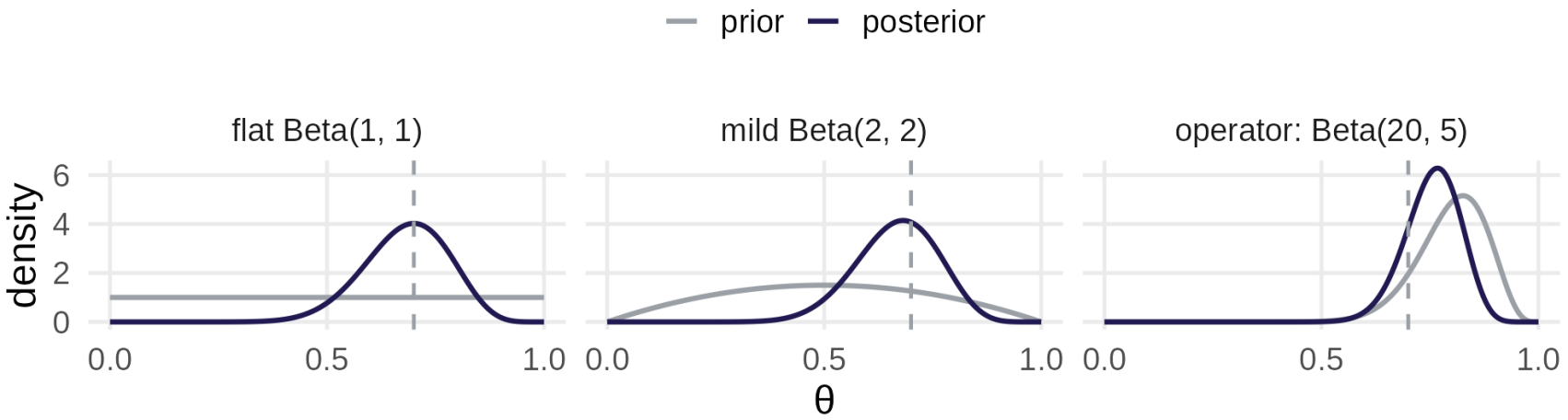
- Neither says that 90% of departures fall in the interval: θ is a probability, not a departure.
- The credible interval is only as good as the model and the prior.

Updating one departure at a time



- Yesterday's posterior is today's prior — as in the marble game.
- After all 20 departures: Beta(16, 8), whatever the order.
- The interval narrows as data arrive.

How much does the prior matter?



prior	posterior	mean	90% interval	P(θ > 0.6)
flat Beta(1, 1)	Beta(15, 7)	0.682	0.51 – 0.83	0.80
mild Beta(2, 2)	Beta(16, 8)	0.667	0.50 – 0.81	0.76
operator says 80%: Beta(20, 5)	Beta(34, 11)	0.756	0.65 – 0.85	0.99

When there is no formula: a grid

```
grid <- tibble(theta = seq(0, 1, length.out = 1001)) |>
  mutate(unnorm = dbeta(theta, 2, 2) * dbinom(14, 20, theta),
         post   = unnorm / sum(unnorm))

grid |> summarise(mean = sum(theta * post))   # 0.667
```

- Works for any prior — try one with no name.
- Read |> as “and then”; mutate() adds columns; summarise() collapses to one row.
- With many parameters a grid becomes impossible: that is where Stan comes in.

First look at Stan

What it is

A language for writing a model: data, parameters, prior and likelihood.

What it gives

Thousands of draws from the posterior — no formula needed.

Why today

We can check Stan against the exact Beta(16, 8). On Day 3 you run it yourselves.

Today it is a demo

Watch, do not install anything now. The demo notebooks are in day-2/demos/ to run after the course.

The bus model in Stan

```
data {
  int<lower=0> n;
  int<lower=0, upper=n> y;
  real<lower=0> a;
  real<lower=0> b;
}
parameters {
  real<lower=0, upper=1> theta;
}
model {
  theta ~ beta(a, b);      // prior
  y ~ binomial(n, theta);  // likelihood
}
```

data

what you know: n, y and the prior's a and b

parameters

what you want to learn: θ, with its range

model

prior and likelihood — the two pieces of Bayes' rule

Running it from R

```
library(cmdstanr)

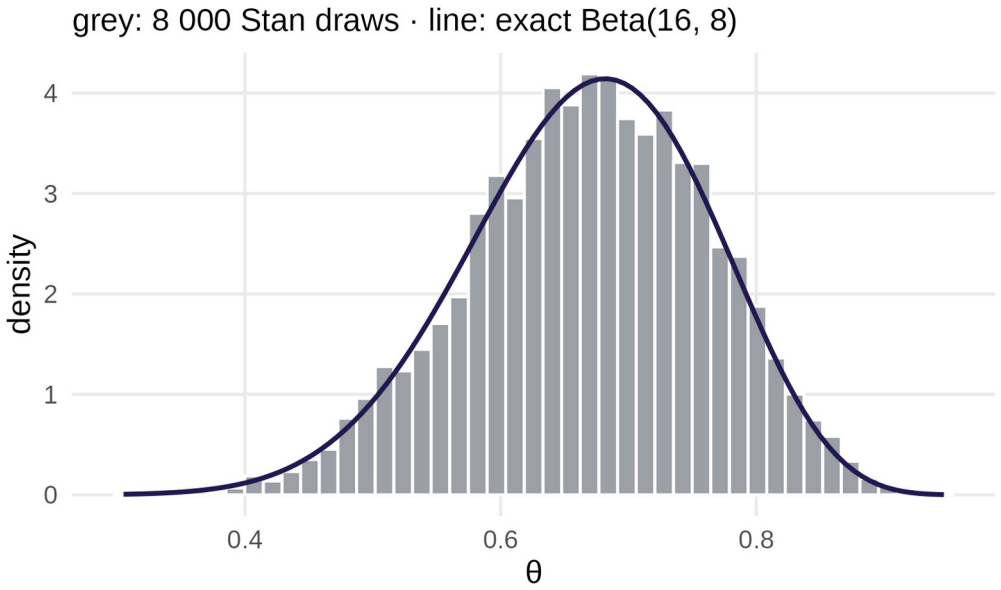
mod <- cmdstan_model("models/beta_binomial.stan")   # compile once

fit <- mod$sample(data = list(n = 20, y = 14, a = 2, b = 2),
                  chains = 4, seed = 2026)

draws <- fit$draws(format = "draws_df")              # one row per draw
draws |> summarise(mean = mean(theta),
                   p_above_6 = mean(theta > 0.6))
```

A posterior probability from draws = the share of draws for which the statement is true.

Stan agrees with the exact answer



	exact	Stan
mean	0.667	0.669
90% interval	0.504 – 0.814	0.504 – 0.818
P(θ > 0.6)	0.763	0.768

Small differences come from using a finite number of draws (Monte Carlo error, Day 3).

Worked example: predict, then compute

1 Strong prior

The operator claims 80% on time: Beta(20, 5). Same data, 14 of 20.

Does the posterior mean go up or down? By how much?

2 Ten times the data

Flat prior, 140 of 200 on time. Same proportion as before.

What happens to the interval? And to the operator's prior?

- Discuss with your neighbour for 2 minutes, then we compute in R.
- Write your guess down first — the surprise is the point.

Worked example: answers

1 Strong prior: Beta(20, 5) + 14 of 20

Posterior Beta(34, 11)

mean 0.756 (up from 0.667) · 90%: 0.65 – 0.85

$P(\theta > 0.6) = 0.99$

The prior counts as 25 departures — more than the 20 we watched.

2 Ten times the data: 140 of 200

prior	posterior	mean	90% interval
Beta(1, 1)	Beta(141, 61)	0.698	0.64 – 0.75
Beta(20, 5)	Beta(160, 65)	0.711	0.66 – 0.76

With 14 of 20 the flat-prior interval was 0.51 – 0.83: ten times the data makes it about 2.5 times narrower, and the two priors now nearly agree.

A Beta(a, b) prior counts as a + b observations: it matters with 20 departures, much less with 200.

Exercise A · 09:45 – 11:10

- Derive one beta–binomial update by hand, then compute it in R.
- Compare two priors: what changes, and why?
- Compute and explain one posterior probability.
- Check your answer with a small grid.
- Write your group's question set: concept · computation · pitfall, with worked solutions.

Everyone back at

11:10

peer review A

Take a break when it suits your group.
Rotate roles: coder · checker · question writers · interviewer · reporter.

Peer review A · 11:10 – 11:55

20 min

Partner interviews

Round 1: odd group asks even group. Round 2: swap. Two questions each, to named members.

23 min

Plenary

Two groups drawn: one interviews the other. Then the reporters' hardest open questions.

2 min

Wrap-up

What to take into lunch and Lecture B.

Partner pairs: 1 ↔ 2 · 3 ↔ 4 · 5 ↔ 6 · 7 ↔ 8 · 9 ↔ 10

```
sample(not_yet_drawn, 2)   # plenary draw
```